Question 1 : What is Tableau? Explain its importance in Business Intelligence and how it helps in data-driven decision-making.
Ans:- **What is Tableau?**

Tableau is a powerful **Business Intelligence (BI)** and **data visualization** tool that helps users analyze, visualize, and share data through interactive dashboards and reports. It enables users to connect to multiple data sources, transform raw data into meaningful insights, and present information using charts, graphs, maps, and dashboards without requiring extensive programming knowledge.

Importance of Tableau in Business Intelligence

Tableau plays a significant role in Business Intelligence because it helps organizations:

1. **Visualize Complex Data**
   - Converts large datasets into easy-to-understand visualizations such as bar charts, line charts, pie charts, heat maps, and geographic maps.
2. **Integrate Multiple Data Sources**
   - Connects to databases, Excel files, cloud platforms, SQL servers, and big data sources, providing a unified view of business data.
3. **Real-Time Data Analysis**
   - Supports live data connections, allowing businesses to monitor performance and make timely decisions.
4. **Interactive Dashboards**
   - Users can filter, drill down, and explore data interactively to gain deeper insights.
5. **Improves Collaboration**
   - Dashboards and reports can be shared securely across teams, enabling better communication and collaboration.
6. **Easy to Use**
   - Its drag-and-drop interface allows both technical and non-technical users to analyze data quickly.

How Tableau Helps in Data-Driven Decision-Making

Tableau supports data-driven decision-making by:

- Identifying trends, patterns, and business opportunities.
- Monitoring Key Performance Indicators (KPIs) in real time.
- Detecting problems and anomalies early through visual analysis.
- Forecasting future performance using historical data.
- Enabling managers to make informed decisions based on facts rather than assumptions.
- Helping businesses optimize operations, improve customer satisfaction, and increase profitability.

Question 2 : Explain the role of the following Tableau components:

a) Data Pane

b) Worksheet

c) Dashboard

d) Story

a) Data Pane

The **Data Pane** is located on the left side of the Tableau workspace and displays all the data connected to the workbook.

Role:

- Shows available tables, fields, and data sources.
- Organizes data into **Dimensions** (categorical data) and **Measures** (numerical data).
- Allows users to drag and drop fields into the worksheet to create visualizations.
- Enables creation of calculated fields, groups, hierarchies, and sets.

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b) Worksheet

A **Worksheet** is the basic workspace in Tableau where users create individual charts, graphs, and visualizations.

Role:

- Used to build a single data visualization.
- Allows users to analyze data using charts, maps, tables, and graphs.
- Supports filtering, sorting, grouping, and formatting.
- Serves as the building block for dashboards and stories.

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c) Dashboard

A **Dashboard** is a collection of multiple worksheets and other objects displayed together on a single screen.

Role:

- Combines different visualizations into one interactive view.
- Helps users compare multiple reports simultaneously.
- Supports filters and actions that update all visualizations together.
- Provides a comprehensive overview of business performance and Key Performance Indicators (KPIs).
- Makes it easier to share insights with stakeholders.

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d) Story

A **Story** is a sequence of worksheets and dashboards arranged to present data in a logical order.

Role:

- Communicates data insights through a step-by-step narrative.
- Consists of multiple **Story Points**, each highlighting a specific insight.
- Helps explain trends, patterns, and business findings.
- Useful for presentations, reports, and decision-making.
- Guides viewers from data analysis to conclusions and recommendations.

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Question 3 : What is the difference between Dimensions and Measures in Tableau? Provide examples of each.
Ans :- In Tableau, **Dimensions** and **Measures** are the two main types of data fields used to create visualizations.

Dimensions

Dimensions are **qualitative (categorical)** fields that describe or categorize data. They are generally used to group, label, filter, or segment data and are displayed as **blue fields** (discrete) by default in Tableau.

Examples of Dimensions:

- Customer Name
- Product Name
- Category
- Sub-Category
- City
- State
- Country
- Region
- Order ID
- Order Date

Example:
If you create a chart showing **Sales by Region**, **Region** is a **Dimension** because it categorizes the sales data.

Measures

Measures are **quantitative (numerical)** fields that can be calculated, aggregated, or analyzed. They are usually displayed as **green fields** (continuous) by default in Tableau.

Examples of Measures:

- Sales
- Profit

- Quantity
- Discount
- Cost
- Revenue
- Average Salary

Question 4 : Define and explain the purpose of Filters, Parameters, and Sets in Tableau.
Ans:- **1. Filters**

Definition

A Filter in Tableau is used to limit the data displayed in a visualization by including or excluding specific records.

Purpose

- Focus on relevant data.
- Remove unnecessary information.
- Improve analysis and dashboard performance.
- Enable interactive exploration of data.

Types of Filters

- Extract Filter
- Data Source Filter
- Context Filter
- Dimension Filter
- Measure Filter

Example

If a sales dashboard contains data for all countries, a filter can be applied to show only India or North Region.

2. Parameters

Definition

A Parameter is a dynamic user-controlled value that can be used in calculations, filters, reference lines, and other Tableau features.

Purpose

- Allow users to change values interactively.
- Make dashboards flexible and dynamic.
- Support what-if analysis and user-driven calculations.

Example

A parameter called Top N Customers can let users choose whether to display the Top 5, Top 10, or Top 20 customers in a chart.

Key Point

Unlike filters, parameters are not directly tied to a specific field and can control multiple parts of a workbook.

3. Sets

Definition

A Set is a custom subset of data created from one or more dimension members.

Purpose

- Group selected members together.
- Compare a subset against the remaining data.
- Perform advanced analysis such as Top/Bottom analysis.

Types of Sets

- Static Set (manually selected members)
- Dynamic Set (based on conditions or rankings)

Example

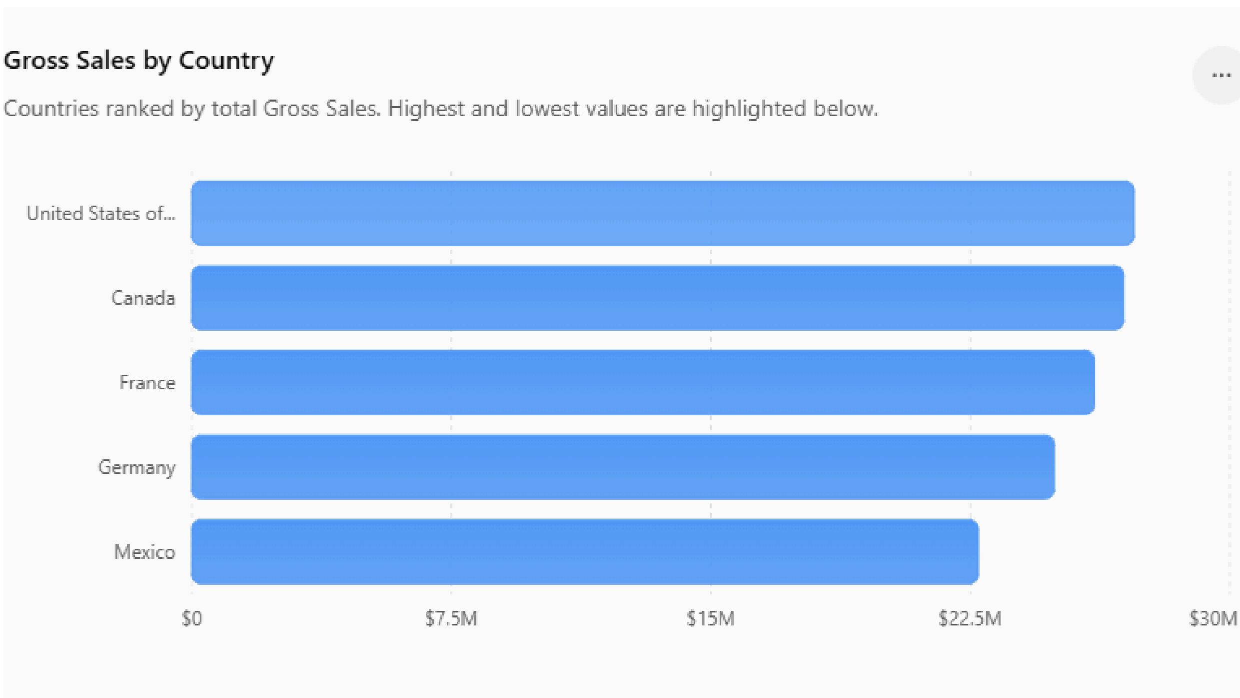
Create a set containing the Top 10 Customers by Sales and compare their performance against All Other Customers.

Question 5 : Create a bar chart showing Gross Sales by Country.
● 🔗 Dataset Link:Global_sales_dataset
● Sort the countries in descending order of sales
● Highlight or annotate the bar that represents the maximum and minimum Gross Sales.
● Add data labels and format the chart for presentation.
Ans:-

Gross Sales by Country (Sorted in Descending Order)

Gross Sales by Country

Countries ranked by total Gross Sales. Highest and lowest values are highlighted below.



Maximum Gross Sales

- **Country:** United States of America
- **Gross Sales: $27,269,358.00**
- **Annotation: ⭐ Maximum Gross Sales**

Minimum Gross Sales

- **Country:** Mexico
- **Gross Sales: $22,726,935.00**
- **Annotation: ⭐ Minimum Gross Sales**

Data Labels

Enable **Show Mark Labels** so each bar displays its Gross Sales value.

Formatting for Presentation

- **Chart Title:** *Gross Sales by Country*
- **Chart Type:** Horizontal Bar Chart
- **Sort:** Descending by Gross Sales
- **Number Format:** Currency ($) with comma separators
- **Fit:** Entire View
- **Font:** 16–18 pt Bold for title; 10–12 pt for labels
- **Highlight:** Color the highest bar (e.g., green) and the lowest bar (e.g., red)
- Remove unnecessary gridlines for a cleaner appearance.

1. Sort Countries in Descending Order

- Click the **Sort Descending** icon or:
  - Right-click **Country** → **Sort**
  - **Sort Order:** Descending
  - **Field:** Gross Sales
  - **Aggregation:** SUM

2. Highlight Maximum Gross Sales

- Select the **United States of America** bar.
- Right-click → **Annotate** → **Mark**
- Annotation:

  Maximum Gross Sales
  $27,269,358.00

- Change the bar color (e.g., green).

3. Highlight Minimum Gross Sales

- Select the **Mexico** bar.
- Right-click → **Annotate** → **Mark**
- Annotation:

 Minimum Gross Sales
$22,726,935.00

- Change the bar color (e.g., red).

Question 6 : Using Tableau, create a dual-axis chart that displays:

● Monthly Sales as bars

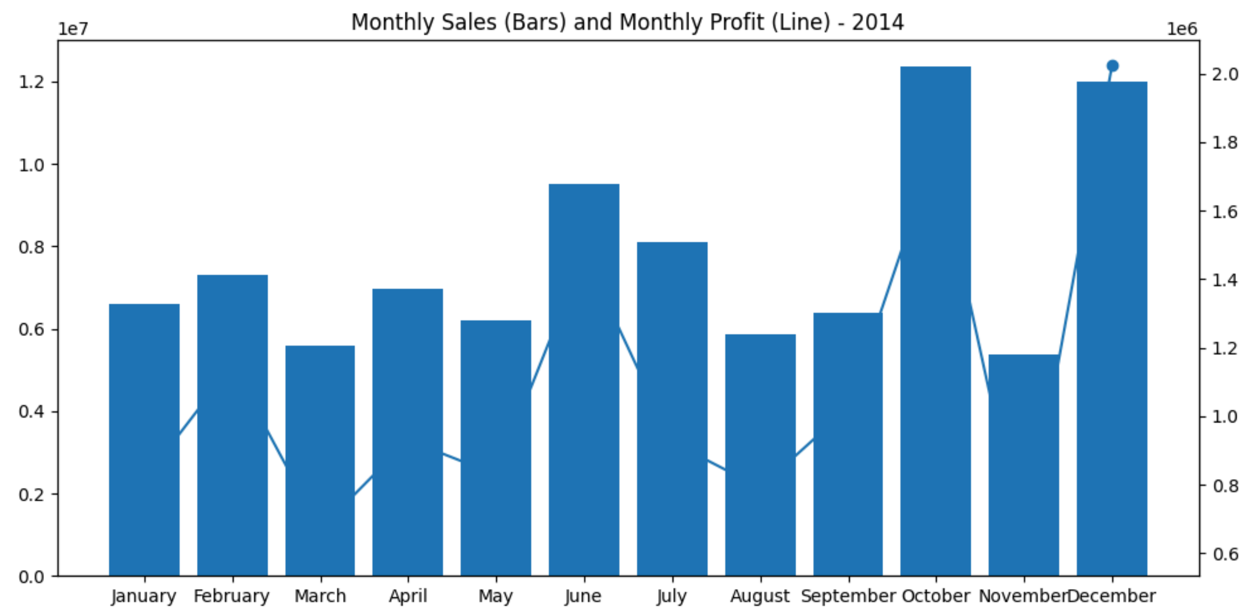
● Monthly Profit as a line

● Filter the data to include only records from the year 2014

● Ensure both axes are synchronized and properly labeled

● Add an appropriate chart title, and format the chart for clear visual presentation
● Paste a screenshot of the final chart in your submission

Ans:-



Question 7 : Create a filled map showing total Units Sold by Country. ●Dataset Link:Global_sales_dataset

● Add a parameter to allow users to switch between Units Sold and Profit.

● Use the Discount Band as a filter in your visualization.

Answer:- Create a **Filled Map** that:

- Displays **Total Units Sold by Country**
- Includes a **Parameter** to switch between **Units Sold** and **Profit**
- Uses **Discount Band** as a filter

Step 1: Connect the Dataset

Open Tableau and connect to the **Global_sales_dataset.xlsx** file.

Step 2: Create the Filled Map

1. Double-click **Country**.
2. Tableau will generate a map automatically.
3. From the **Marks** card, change the mark type to **Filled Map**.

Step 3: Create the Parameter

Go to **Data Pane → Right-click → Create Parameter**

Name:

Measure Selector

Data Type:

String

Allowable Values:

List

Add two values:

Value	Display As
Units Sold	Units Sold
Profit	Profit

Click **OK**.

Right-click the parameter and select

Show Parameter

Step 4: Create a Calculated Field

Create a calculated field named

Selected Measure

Formula:

IF [Measure Selector] = "Units Sold" THEN

  SUM([Units Sold])

ELSE

  SUM([Profit])

END

Click **OK**.

Step 5: Color the Map

Drag

Selected Measure

onto **Color**.

Now the map changes based on the selected parameter.

Step 6: Add Discount Band Filter

Drag

Discount Band

to the **Filters** shelf.

Choose

All

Click **OK**.

Right-click the filter →

Show Filter

Users can now select:

- None
- Low
- Medium
- High

Step 7: Format

Title:

Total Units Sold / Profit by Country

Choose a sequential color palette.

Show Country labels if required.

Display both:

- Measure Selector (Parameter)
- Discount Band Filter

on the right side.

Question 8 : Create a dashboard that includes: ● 🔗 Dataset Link:Global_sales_dataset ● KPI tiles for Total Sales, Total Profit, and Total Units Sold ● A line chart for Profit trend over time ● Filters for Product and Country Ensure your dashboard is interactive and visually appealing

Ans:- **Step 1: Connect to the Dataset**

Open **Tableau Desktop** and connect to **Global_sales_dataset.xlsx**.

Step 2: Create KPI Sheet – Total Sales

1. Create a new worksheet.
2. Drag **Sales** to **Text** on the Marks card.
3. Change aggregation to **SUM**.
4. Increase font size (24–36 pt).
5. Rename the sheet:

Total Sales

Step 3: Create KPI Sheet – Total Profit

Create another worksheet.

Drag

Profit

to **Text**.

Aggregation:

SUM(Profit)

Rename sheet:

Total Profit

Step 4: Create KPI Sheet – Total Units Sold

Create another worksheet.

Drag

Units Sold

to **Text**.

Aggregation:

SUM(Units Sold)

Rename sheet:

Total Units Sold

Step 5: Create Profit Trend Line Chart

Create a new worksheet.

1. Drag **Date** (or Order Date) to **Columns**.
2. Change to **Month** (Continuous).
3. Drag **Profit** to **Rows**.
4. Marks → **Line**.
5. Rename the sheet:

Monthly Profit Trend

Step 6: Create Filters

Drag these fields onto the Filters shelf:

- Product
- Country

Right-click each field and choose:

Show Filter

Step 7: Build the Dashboard

Go to

Dashboard → New Dashboard

Recommended dashboard size:

Automatic

or

1200 × 800

Arrange the sheets like this:

```
--------------------------------------------------------

         GLOBAL SALES DASHBOARD

--------------------------------------------------------


+---------------+---------------+----------------+

| Total Sales   | Total Profit  | Units Sold     |

+---------------+---------------+----------------+


+-----------------------------------------------------+

|                                             |

|           Monthly Profit Trend (Line Chart)          |

|                                             |

+-----------------------------------------------------+


Product Filter


Country Filter


--------------------------------------------------------
```

Step 8: Make the Dashboard Interactive

From the Dashboard menu:

Dashboard

→ Actions

→ Add Action

→ Filter

Or simply click the filter cards and choose:

Apply to Worksheets

→ All Using This Data Source

This allows:

- Selecting a **Product** updates KPIs and chart.
- Selecting a **Country** updates KPIs and chart.

Step 9: Format the Dashboard

- Add the title:

Global Sales Dashboard

- Use consistent fonts (Calibri or Tableau Book).
- Format Sales and Profit as Currency.
- Format Units Sold as Whole Number.
- Remove unnecessary gridlines.
- Use contrasting colors for the line chart.
- Add borders or shading to KPI tiles for better visibility.

Final Dashboard Layout

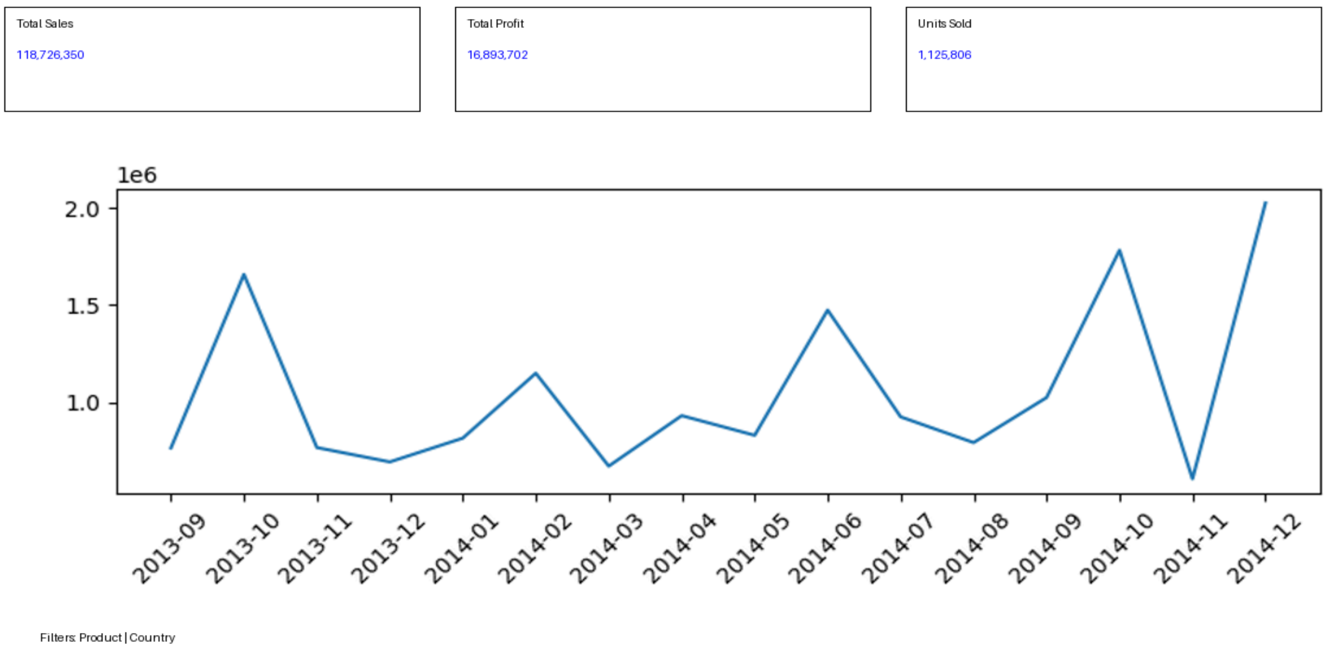
```
==================================================================
              GLOBAL SALES DASHBOARD
==================================================================

┌───────────────────┬───────────────────┬─────────────────────┐
│ Total Sales │ Total Profit │ Total Units  │
│   $XXX,XXX  │   $XX,XXX   │    XX,XXX    │
└───────────────────┴───────────────────┴─────────────────────┘

┌─────────────────────────────────────────────────────────────
────────────┐
│                              │
│        Monthly Profit Trend (Line Chart)         │
│                              │
└─────────────────────────────────────────────────────────────
────────────┘

Product Filter    Country Filter

==================================================================
```



Question 9 : Your goal is to identify products that generate low profit despite high sales volume. ● 🔗 Dataset Link:Global_sales_dataset ● Use scatter plot or highlight table to identify such products. ● Add filters for Country and Segment. ● Write two business insights based on your chart.

Ans:- **Step 1: Connect the Dataset**

Open **Tableau Desktop** and connect to **Global_sales_dataset.xlsx**.

Step 2: Create the Scatter Plot

1. Open a new worksheet.

2. Drag **Sales** to the **Columns** shelf.
3. Drag **Profit** to the **Rows** shelf.
4. Drag **Product** to **Detail** on the Marks card.
5. Drag **Units Sold** to **Size**.
6. Drag **Product** to **Label** (optional, for easier identification).
7. Set the Marks type to **Circle**.

Your scatter plot will show:

- **X-axis:** Sales
- **Y-axis:** Profit
- **Bubble Size:** Units Sold
- **Each Point:** Product

Products in the **bottom-right area** (high Sales, low Profit) are the ones of interest.

Step 3: Add Filters

Drag the following fields to the **Filters** shelf:

- **Country**
- **Segment**

Right-click each field and select:

Show Filter

Users can now interactively filter the scatter plot.

Step 4: Format the Chart

Chart Title:

Products with High Sales Volume but Low Profit

Format:

- Sales as Currency
- Profit as Currency
- Display product labels for outliers
- Use color if desired to distinguish product categories

Expected Layout

Products with High Sales Volume but Low Profit

```
Profit
^
|
|                   ●
|               ●
|           ●
|       ●
|  ●
|____________________________________________________________> Sales
```

(Bubble size = Units Sold)

Filters:
Country ▼

Segment ▼

Business Insights

1. **Several products generate high sales but relatively low profit, indicating that strong sales volume does not always translate into profitability. These products may have high discounts, production costs, or low profit margins.**

2. **Comparing results by Country and Segment helps identify markets where profitability is weaker despite good sales performance. This enables managers to review pricing strategies, discount policies, and product mix to improve overall profit.**